Midterm Formula Sheet

This sheet is printed and handed out with the exam. You do not need to memorize these formulas — you need to know what they mean and how to use them.

Linear regression (one feature)

- Model (prediction): $\hat{y} = f_{(w,b)}(x) = w \cdot x + b$
- Error for example *i*: $e^{(i)} = f_{(w,b)}(x^{(i)}) - y^{(i)}$ (prediction minus true value)
- Cost (mean squared error, halved): $J(w, b) = (1 / 2m) \cdot \Sigma_i \, (f_{(w,b)}(x^{(i)}) - y^{(i)})^2$, where *m* = number of examples

Gradient descent

- Gradients: $\partial J/\partial w = (1/m) \cdot \Sigma_i \, (f_{(w,b)}(x^{(i)}) - y^{(i)}) \cdot x^{(i)}$ $\partial J/\partial b = (1/m) \cdot \Sigma_i \, (f_{(w,b)}(x^{(i)}) - y^{(i)})$
- Update (both at the same time): $w \leftarrow w - \alpha \cdot \partial J/\partial w$ $b \leftarrow b - \alpha \cdot \partial J/\partial b$ (α = learning rate)

Evaluation (regression)

- Mean absolute error: $MAE = (1/m) \cdot \Sigma \, |\hat{y}^{(i)} - y^{(i)}|$
- Root mean squared error: $RMSE = \sqrt{[\, (1/m) \cdot \Sigma \, (\hat{y}^{(i)} - y^{(i)})^2 \,]}$
- Coefficient of determination: $R^2 = 1 - \Sigma \, (y^{(i)} - \hat{y}^{(i)})^2 \, / \, \Sigma \, (y^{(i)} - \bar{y})^2$ ($\bar{y}$ = mean of y)

Neural networks

- One neuron: $z = w_1 x_1 + w_2 x_2 + \ldots + b$, then output $a = g(z)$
- Activation functions: $ReLU(z) = \max(0, z)$ $sigmoid(z) = 1 / (1 + e^{-z})$
- Parameters in one fully connected layer with *n_in* inputs and *n_out* neurons: $n_in \times n_out + n_out$ (weights + biases)

Evaluation (classification)

	Predicted positive	**Predicted negative**
Actually positive	True positive (TP)	False negative (FN)
Actually negative	False positive (FP)	True negative (TN)

- **Accuracy = (TP + TN) / (TP + TN + FP + FN)**
- **Precision = TP / (TP + FP)** — of the items flagged positive, the fraction that really are
- **Recall = TP / (TP + FN)** — of the real positives, the fraction we found